

hovid Dexamethasone Phosphate

Injection 4 mg / mL

VIDEXx-PC (FEO)

DESCRIPTION

Transparent solution, colourless or slightly yellowish in colour.

COMPOSITION

Each 1 mL ampoule contains dexamethasone sodium phosphate equivalent to 4 mg dexamethasone phosphate or 3.33 mg dexamethasone.

ACTIONS AND PHARMACOLOGY

Pharmacology of the corticosteroids is complex and the drugs affect almost all body systems. Maximum pharmacological activity lags behind peak blood concentrations, suggesting that most effects of the drugs result from modification of enzyme activity rather than from direct actions by the drugs.

PHARMACOKINETICS

Intramuscular injections of dexamethasone phosphate gives maximum plasma concentrations of dexamethasone at 1 hour. Binding of dexamethasone to plasma proteins is less than for most other corticosteroids. Dexamethasone penetrates into tissue fluids and cerebrospinal fluids. Metabolism of the drug takes place in the kidneys and liver and excretion is via the urine.

INDICATION

Replacement therapy - adrenocortical insufficiency

Dexamethasone has predominantly glucocorticoid activity and therefore is not a complete replacement therapy in cases of adrenocortical insufficiency. Dexamethasone should be supplemented with salt and/or a mineralocorticoid, such as deoxycorticosterone. When so supplemented, dexamethasone is indicated in:

- Acute adrenocortical insufficiency - Addison's disease, bilateral adrenalectomy.
- Relative adrenocortical insufficiency - Prolonged administration of adrenocortical steroids can produce dormancy of the adrenal cortex. The reduced secretory capacity gives rise to a state of relative adrenocortical insufficiency which persists for a varying length of time after therapy is discontinued. Should a patient be subjected to sudden stress during this period of reduced secretion (for up to two years after therapy has ceased) the steroid output may not be adequate. Steroid therapy should therefore be reinstituted to help cope with stress such as that associated with surgery, trauma, burns, or severe infections where specific antibiotic therapy is available.
- Primary and secondary adrenocortical insufficiency

Disease therapy

Dexamethasone is indicated for therapy of the following diseases:

Collagen diseases: Systemic lupus erythematosus, polyarteritis nodosa, dermatomyositis, giant cell arteritis, adjunctive therapy for short term administration during an acute episode or exacerbation, acute rheumatic carditis - during an exacerbation or as maintenance therapy.

Pulmonary disorders: Status asthmaticus, chronic asthma, sarcoidosis, respiratory insufficiency.

Blood disorders: Leukaemia, idiopathic thrombocytopenic purpura in adults, acquired (autoimmune) haemolytic anaemia.

Rheumatic diseases: Rheumatoid arthritis, osteoarthritis, adjunctive therapy for short term administration during an acute episode or exacerbation of rheumatoid arthritis or osteoarthritis.

Skin diseases: Psoriasis, erythema multiforme, pemphigus, neutrophilic dermatitis, localised neurodermatitis, exfoliative dermatitis, sarcoidosis skin, severe seborrhoeic dermatitis, contact dermatitis.

Gastrointestinal disorders: Ulcerative colitis, regional enteritis.

Oedema: Cerebral oedema associated with primary or metastatic brain tumours, neurosurgery, or stroke, oedema associated with acute noninfectious laryngospasm (or laryngitis).

Eye disorders: Allergic conjunctivitis, keratitis, allergic corneal marginal ulcers, choroiditis, optic neuritis, anterior ischaemic optic neuropathy.

Neoplastic states: Cerebral neoplasms, hypercalcaemia associated with cancer, leukaemia and lymphomas in adults, acute leukaemia in children.

Endocrine disorders: Adrenal insufficiency.

Preoperative and postoperative support

Dexamethasone may be used in any surgical procedure when the adrenocortical reserve is doubtful. This includes the treatment of shock due to excessive blood loss during surgery.

Shock

Dexamethasone may be used as an adjunct in the treatment of shock. Vials of Dexamethasone Phosphate 120 milligrams/5 mL are specifically designed for shock therapy where high pharmacological doses may be needed. Dexamethasone should not be used as a substitute for normal shock therapy.

CONTRAINDICATIONS

Unless considered to be life-saving systemic administration of corticosteroids are generally contraindicated in patients with systemic infections, (unless specific anti-infective therapy is employed). Hypersensitivity to any components of the injection.

Administration of dexamethasone is contraindicated in patients with a known hypersensitivity to sulphites.

WARNINGS AND PRECAUTIONS

- Dexamethasone formulations containing sulphites should not be used for intrathecal therapy. This sulphite-containing formulation has an altered risk profile compared to other sulphite-free formulations; there is a potential risk of neurotoxicity when administered intrathecally.
- The excipient sodium sulphite may rarely cause severe hypersensitivity reactions and bronchospasm.
- Patients and/or carers should be warned that potentially severe psychiatric adverse reactions may occur with systemic steroids. Symptoms typically emerge within a few days or weeks of starting the treatment. Risks may be higher with high doses/systemic exposure, although dose levels do not allow prediction of the onset, type, severity or duration of reactions. Most reactions recover after either dose reduction or withdrawal, although specific treatment may be necessary. Patients/carers should be encouraged to seek medical advice if worrying psychological symptoms develop, especially if depressed mood or suicidal ideation is suspected. Patients/carers should also be alert to possible psychiatric disturbances that may occur either during or immediately after dose tapering/withdrawal of systemic steroids, although such interactions have been reported infrequently.
- Particular care is required when considering the use of systemic corticosteroids in patients with existing or previous history of severe affective disorders in themselves or in their first degree relatives. These would include depressive or manic-depressive illness and previous steroid psychosis.
- The lowest effective dose of corticosteroid should be used to control the condition under treatment for the minimum period. Frequent patient review is required to appropriately titrate the dose against disease activity. When dose reduction is possible, it should occur gradually. Too rapid a reduction of dexamethasone dosage following prolonged treatment can lead to acute adrenal insufficiency, hypotension and death.
- A 'withdrawal syndrome' may also occur including fever, myalgia, arthralgia, rhinitis, conjunctivitis, painful itchy skin nodules and loss of weight.

- Adrenal suppression: Adrenal cortical atrophy develops during prolonged therapy and may persist for years after stopping treatment. Withdrawal of corticosteroids after prolonged therapy must, therefore, be gradual to avoid acute adrenal insufficiency, being tapered off over weeks or months according to the dose and duration of treatment. During prolonged therapy any intercurrent illness, trauma or surgical procedure will require a temporary increase in dosage; if corticosteroids have been stopped following prolonged therapy they may need to be temporarily re-introduced.
- There is a lack of evidence to support the prolonged use of corticosteroids in septic shock.
- Severe anaphylactoid reactions have occurred after administration of parenteral corticosteroids, particularly in patients with history of allergy. Appropriate precautions should be taken prior to administration.
- The slower rate of absorption after intramuscular injection should be noted.
- Intraarticular corticosteroids are associated with a substantially increased risk of an inflammatory response in the joint, particularly a bacterial infection introduced with the injection. Great care is required and all intraarticular corticosteroid injections should be undertaken in an aseptic environment. Charcot like arthropathies have been reported particularly after repeated injections.
- Prior to intraarticular injection the joint fluid should be examined to exclude a septic process. A marked increase in pain, accompanied by local swelling, fever, restriction of joint motion, fever and malaise are suggestive of septic arthritis. If this complication occurs and sepsis is confirmed, appropriate antimicrobial therapy should be commenced.
- Patients should be impressed strongly with the importance of not overusing joints in which symptomatic benefit has been obtained, but the inflammatory process remains active.
- Suppression of the inflammatory response and the immune function increases the susceptibility to infections and their severity. The clinical presentation may be atypical and serious infections, such as septicæmia and tuberculosis, may be masked and may reach an advanced stage before being recognised.
 - Chickenpox is of particular concern since this normally minor illness may be fatal in immunosuppressed patients. Patients (or parents of children) without a definite history of chickenpox should be advised to avoid close personal contact with chickenpox or herpes zoster and if exposed they should seek urgent medical attention. Passive immunisation with varicella zoster immunoglobulin (VZIG) is needed by exposed non-immune patients who are receiving systemic dexamethasone or who have received it during the previous 3 months; this should be given within 10 days of exposure to chickenpox. If a diagnosis of chickenpox is confirmed, the illness warrants specialist care and urgent treatment. Dexamethasone should not be stopped and the dose may need to be increased.
 - Live vaccines should not be given to individuals with impaired immune responsiveness. The antibody response to other vaccines may be diminished.
 - False negative results may occur with the nitroblue tetrazolium test for bacterial infection.
- Extreme caution should be exercised in the treatment of patients with the following conditions and frequent patient monitoring is necessary:
 - Liver failure, chronic renal failure, congestive heart failure, hypertension, epilepsy, migraine.
 - Osteoporosis, since corticosteroids increase calcium excretion. Post-menopausal women are at particular risk.
 - Latent tuberculosis, as corticosteroids can cause reactivation.
 - Hypothyroidism or cirrhosis, because such patients often show an exaggerated response to corticosteroids.
 - Latent amoebiasis, as corticosteroids may cause reactivation. Prior to treatment, amoebiasis should be ruled out in any patient with unexplained diarrhoea or who has recently spent time in the tropics.
 - Ocular herpes simplex, because corticosteroids may cause corneal perforation.
- Corticosteroids should also be used with caution in patients with diabetes mellitus (or a family history of diabetes), affective disorders (especially previous steroid psychosis), glaucoma (or a family history of glaucoma), peptic ulceration or previous corticosteroid-induced myopathy.

PREGNANCY AND LACTATION

Pregnancy: The ability of corticosteroids to cross the placenta varies between individual drugs, however, dexamethasone readily crosses the placenta.

As with all drugs, corticosteroids should only be prescribed when the benefits to the mother and child outweigh the risks. When corticosteroids are essential however, patients with normal pregnancies may be treated as though they were in the non-gravid state.

There is evidence of harmful effects on pregnancy in animals. Infants born to mothers who have received substantial doses of corticosteroids during the pregnancy should be carefully observed, for signs of adrenal insufficiency.

Patients with pre-eclampsia or fluid retention require close monitoring.

Lactation: Corticosteroids may pass into breast milk, although no data are available for dexamethasone. Infants of mothers taking high doses of systemic corticosteroids for prolonged periods may have a degree of adrenal suppression. Suppression of growth or other adverse effects may occur.

DRUG INTERACTIONS

Aspirin should be used cautiously in conjunction with corticosteroids in hypoprothrombinaemia.

The renal clearance of salicylates is increased by corticosteroids and therefore salicylate dosage should be reduced along with steroid withdrawal.

As phenytoin, barbiturates, ephedrine, rifabutin, carbamazepine, rifampicin and aminoglutethimide may enhance the metabolic clearance of corticosteroids, resulting in decreased blood levels and reduced physiological activity, the dosage may have to be adjusted. These interactions interfere with dexamethasone suppression tests which should be interpreted with caution during administration of these drugs.

False-negative results in the dexamethasone suppression test in patients being treated with indometacin have been reported.

The efficacy of coumarin anticoagulants may be changed by concurrent corticosteroid treatment. The prothrombin time should be checked frequently in patients who are receiving corticosteroids and coumarin anticoagulants at the same time, in order to avoid spontaneous bleeding.

The desired effects of hypoglycaemic agents (including insulin) are antagonised by corticosteroids.

When corticosteroids are administered concomitantly with potassium-depleting diuretics, patients should be observed closely for development of hypokalaemia.

Corticosteroids may affect the nitroblue tetrazolium test for bacterial infection and produce false-negative results.

Antiretroviral protease inhibitors (ritonavir, darunavir, indinavir, lopinavir, saquinavir and efavirenz) are metabolised by CYP3A. Medicinal products that induce CYP3A activity, such as dexamethasone, may increase the clearance of medicines metabolised by CYP3A, resulting in lowered plasma concentrations.

Certain antiretroviral protease inhibitors (ritonavir, indinavir) may also be inhibitors of CYP3A themselves and as a result may increase the plasma concentration of dexamethasone.

MAIN SIDE/ ADVERSE EFFECTS

A wide range of psychiatric reactions including affective disorders (such as irritable, euphoric, depressed and labile mood, and suicidal thoughts). Psychotic reactions (including mania, delusions, hallucinations and aggravation of schizophrenia), behavioural disturbances, irritability, anxiety, sleep disturbances and cognitive dysfunction including confusion and amnesia have been reported. Reactions are common and may occur in both adults and children. Psychological effects have been reported on withdrawal of corticosteroids; the frequency is unknown.

The incidence of predictable undesirable effects, including hypothalamic-pituitary-adrenal suppression correlates with the relative potency of the drug, dosage, timing of administration and the duration of treatment.

High doses of dexamethasone sodium phosphate are intended for short term therapy and therefore adverse reactions are uncommon. However, peptic ulceration and bronchospasm may occur.

Except for hypersensitivity, the following adverse effects have been associated with prolonged systemic corticosteroid therapy.

Endocrine and metabolic disturbances: Suppression of the hypothalamic-pituitary-adrenal axis; Cushing-like syndrome, hirsutism and weight gain; suppression of growth in infants, children and adolescents; secondary adrenocortical unresponsiveness, particularly in times of stress, as in surgery or trauma; menstrual irregularities and amenorrhoea; impaired glucose tolerance with increased requirement for anti-diabetic therapy; hyperglycaemia; negative protein/nitrogen and calcium balance; increased appetite.

Metabolic: Electrolyte imbalance (retention of sodium and water with oedema and hypertension); nitrogen depletion; hyperglycaemia; hypokalaemic alkalosis; increased calcium and potassium excretion and hypertension.

Anti-inflammatory and immunosuppressive effects: Increased susceptibility to and severity of infection with suppression of clinical symptoms and signs; opportunistic infections; recurrence of dormant tuberculosis.

Musculoskeletal: Muscular atrophy, proximal myopathy, premature epiphyseal closure, osteoporosis, avascular osteonecrosis, muscle weakness, tendon rupture, vertebral compression and long bone fractures.

Gastro-intestinal: Dyspepsia, peptic ulceration with perforation and haemorrhage, oesophageal ulcerations, acute pancreatitis and candidiasis.

Dermatological: Impaired wound healing; skin atrophy; bruising; telangiectasia and striae; petechiae and ecchymoses; erythema; increased sweating; possible suppression of skin tests; burning or tingling; bruising; allergic dermatitis; urticaria, candidiasis, acne.

Neurological: Mental disturbances, psychological dependence, euphoria, depression, insomnia, headache, convulsions, vertigo. Aggravation of epilepsy and schizophrenia. Increased intra-cranial pressure with papilloedema in children (pseudotumour cerebri), usually after treatment withdrawal.

Ophthalmic: Posterior sub-capsular cataracts or increased intraocular pressure may result in glaucoma or occasionally damage to the optic nerve; exophthalmos papilloedema; corneal or scleral thinning; exacerbation of ophthalmic viral or fungal diseases.

Others: Hypersensitivity including anaphylaxis, has been reported; illness associated with intraslesional therapy around the face and neck; hyperpigmentation; hypopigmentation; subcutaneous and cutaneous atrophy; sterile abscess; post injection flare (following intraarticular injection); Charcot-like arthropathy, leucocytosis, thromboembolism.

Withdrawal:

In patients who have received more than physiological doses of systemic corticosteroids (approximately 1 mg dexamethasone) for greater than 3 weeks, withdrawal should not be abrupt. How dose reduction should be carried out depends largely on whether the disease is likely to relapse as the dose of systemic corticosteroids is reduced. Clinical assessment of disease activity may be needed during withdrawal. If the disease is unlikely to relapse on withdrawal of systemic corticosteroids but there is uncertainty about HPA suppression, the dose of systemic corticosteroid may be reduced rapidly to physiological doses. Once a daily dose of 1 mg dexamethasone is reached, dose reduction should be slower to allow the HPA-axis to recover.

Abrupt withdrawal of systemic corticosteroid treatment, which has continued up to 3 weeks is appropriate if it is considered that the disease is unlikely to relapse. Abrupt withdrawal of doses up to 6 mg of dexamethasone for 3 weeks is unlikely to lead to clinically relevant HPA-axis suppression, in the majority of patients. In the following patient groups, gradual withdrawal of systemic corticosteroid therapy should be considered even after courses lasting 3 weeks or less:

- Patients who have had repeated courses of systemic corticosteroids, particularly if taken for greater than 3 weeks,
- When a short course has been prescribed within one year of cessation of long-term therapy (months or years),
- Patients who may have reasons for adrenocortical insufficiency other than exogenous corticosteroid therapy,
- Patients receiving doses of systemic corticosteroid greater than 6 mg daily of dexamethasone,
- Patients repeatedly taking doses in the evening.

Withdrawal symptoms and signs: Too rapid a reduction of corticosteroid dosage following prolonged treatment can lead to acute adrenal insufficiency, hypotension and death.

A 'withdrawal syndrome' may also occur including fever, myalgia, arthralgia, rhinitis, conjunctivitis, painful itchy skin nodules and loss of weight.

OVERDOSE AND TREATMENT

Treat anaphylaxis with adrenaline and positive pressure ventilation. Other supportive measures aimed to maintain the patient unstressed.

DOSAGE AND ADMINISTRATION

Dosage of dexamethasone sodium phosphate is usually expressed in terms of dexamethasone phosphate.

Contains no antimicrobial agent. For single patient use. Use once only and discard any residue.

Intravenous and Intramuscular administration

Dosage requirements are variable and must be individualised on the basis of the disease being treated and patient response.

Intravenous or intramuscular dosage usually ranges from 0.5 to 24 milligrams of dexamethasone phosphate daily. The duration of therapy is dependent on the clinical response of the patient and as soon as improvement is indicated, the dosage should be adjusted to the minimum required to maintain the desired clinical response. Withdrawal of the drug on completion of therapy should be gradual.

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Parenteral dexamethasone is generally reserved for patients who are unable to take the drug orally, or for use in an emergency situation.

Shock (of haemorrhagic, traumatic or surgical origin)

The usual dose for the treatment of shock is 2 to 6 milligrams/kg bodyweight as a single intravenous injection. This may be repeated in 2 to 6 hours if shock persists.

An alternative regimen of 20 milligrams by intravenous injection initially, followed by continuous intravenous infusion of 3 milligrams/kg bodyweight per 24 hours, has been suggested. If required for intravenous infusion, dexamethasone phosphate 120 milligrams/5 mL may be diluted with glucose or sodium chloride injection.

High-dose therapy should be continued only until the patient's condition has stabilised and usually for no longer than 48 to 72 hours.

To avoid microbial contamination hazards, infusion should be commenced as soon as practicable after preparation of the mixture. Infusion should be completed within 24 hours of preparation of the solution and any residue discarded.

WARNING: Further diluted solutions which are not clear, or which show evidence of particulate matter contamination, should be discarded.

Cerebral oedema:

The treatment schedule and route of administration should reflect the severity and aetiology of the cerebral oedema. Treatment needs to be tailored to the individual response. An initial dose of 10 milligrams intravenously followed by 4 milligrams intramuscularly every 6 hours until the symptoms of oedema subside (usually after 12 to 24 hours). After 2 to 4 days the dosage should be reduced and gradually stopped over a period of 5 to 7 days. Patients with cerebral malignancy may require maintenance therapy with doses of 2 milligrams intramuscularly or intravenously 2 to 3 times daily.

High doses of dexamethasone may be used to initiate short-term intensive therapy for acute cerebral oedema. Following an initial high loading dose, the dose is scaled down over the 7 to 10 day period of intensive therapy, and subsequently reduced to zero over the next 7 to 10 days.

High Dose Schedule:

	Adults	Children > 35 kg	Children < 35 kg
Initial dose	50 mg IV	25 mg IV	20 mg IV
1 st day	8 mg IV every 2 hrs	4 mg IV every 2 hrs	4 mg IV every 3 hrs
2 nd day	8 mg IV every 2 hrs	4 mg IV every 2 hrs	4 mg IV every 3 hrs
3 rd day	8 mg IV every 2 hrs	4 mg IV every 2 hrs	4 mg IV every 3 hrs
4 th day	4 mg IV every 2 hrs	4 mg IV every 4 hrs	4 mg IV every 6 hrs
5 th -8 th day	4 mg IV every 4 hrs	4 mg IV every 6 hrs	2 mg IV every 6 hrs
After 8 days	Decrease by daily reduction of 4 mg	Decrease by daily reduction of 2 mg	Decrease by daily reduction of 1 mg

Note: The intravenous and intramuscular routes of administration of hovid Dexamethasone Injection should only be used where acute illness or life-threatening situations exist. Oral therapy should be substituted as soon as possible.

Intra-synovial & soft tissue injections:

Dosage varies with the degree of inflammation and size and location of the affected area. Injections may be repeated from once every 3 to 5 days (eg. for bursae) to once every 2 to 3 weeks (for joints). Frequent intra-articular injection may result in damage to joint tissues.

Site of Injection	Dosage
1. Large Joint	2 mg to 4 mg
2. Small Joints	80 µg to 1 mg
3. Bursae	2 mg to 3mg
4. Tendon Sheaths	400 µg to 1 mg
5. Soft Tissue Infiltration	2 mg to 6 mg
6. Ganglia	1 mg to 2 mg

PREPARATION AND DIRECTIONS FOR USE

When dexamethasone sodium phosphate is given by intravenous infusion, only Sodium Chloride Injection or Glucose Injection should be used as diluents. The exact concentration of dexamethasone per infusion container should be determined by the desired dose, patient fluid intake and drip rate required.

Chemical and physical in-use stability for a prepared solution for infusion using Sodium Chloride Injection or Glucose Injection have been demonstrated for 24 hours at controlled room temperature (2°C-8°C).

- The product should only be used when the solution is clear and particle free.
- Each vial is for single use only.
- Discard any unused portion immediately after use.

Incompatibilities: Dexamethasone sodium phosphate is physically incompatible with daunorubicin, doxorubicin and vancomycin and should not be admixed with solutions containing these drugs. Also incompatible with doxapram HCl and glycopyrrolate in syringe.

Note: The information given here is limited. For further information, consult your doctor or pharmacist.

Storage:

Store below 25°C. Protect from light and freezing.

Presentation/ Packing:

10 ampoules of 1 mL

Product Registration Holder/ Repacked by:HOVID Bhd.

121, Jalan Tunku Abdul Rahman, 30010 Ipoh, Malaysia.
Manufactured by: MJ Biopharm Pvt. Ltd., Plot No L7, MIDC Industrial Area, Talaja, Raigad 410208 Maharashtra State, India.

Information date: July 2016

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